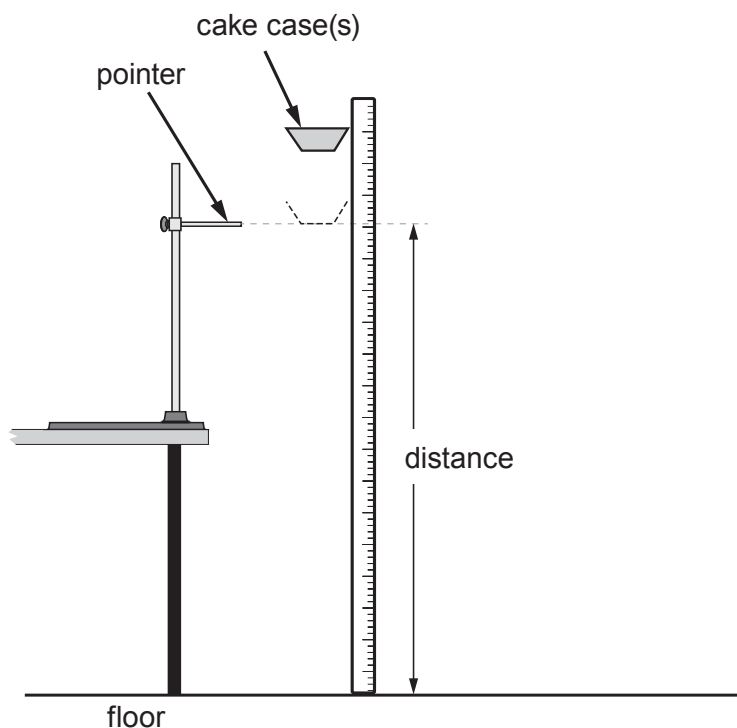


2. A group of students investigate if the terminal speed of falling paper cake cases depends on their mass. They follow the method given below.



1. Set up a pointer in the clamp stand and set it 1.50 m above the ground.
2. Take a single cake case and record its mass using a balance.
3. Drop the cake case from 20 cm above the pointer.
4. Use a stopwatch to record the time it takes to fall from the level of the pointer to the floor.
5. Repeat steps 3 and 4 another four times.
6. Repeat steps 3-5 with extra cake cases in a stack.

- (a) (i) Give **one** reason why the students let the cake case fall for 20 cm before starting the stopwatch. [1]

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- (ii) Give **two** reasons why the students take more than one repeat reading. [2]

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- (b) The following data are collected. The students assume that each of the cake cases has a mass of 0.5g.

Number of cake cases	Mass of cake cases (g)	Mean time taken for cake cases to fall 1.50 m (s)	Terminal speed (m/s)
0	0		0
1	0.5	0.94	1.60
2	1.0	0.67	2.24
3	1.5	0.59	2.54
5	2.5		2.88
6	3.0	0.51	2.94

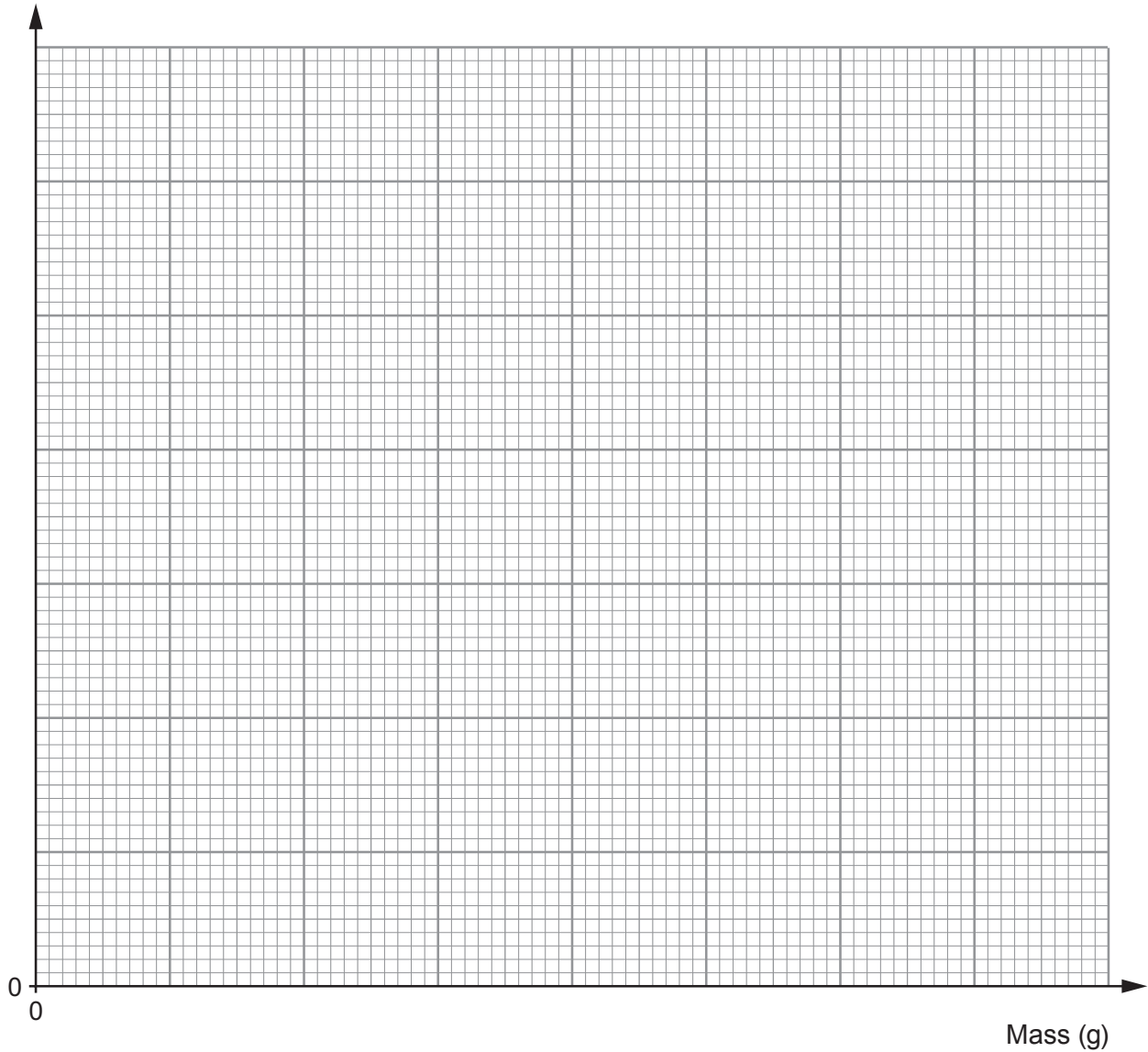
- (i) **Complete the table.** Use an equation from page 2 to calculate the missing value of the mean time. *Space for workings.* [2]



(ii) Plot the data on the grid below and draw a suitable line.

[4]

Terminal speed (m/s)



(iii) One of the students suggests that the terminal speed will always increase by a factor of 1.4 if the mass is doubled. The student finds that when the mass doubles from 0.5g to 1.0g this suggestion is true. Explain if this is true for the other masses. [3]

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- (c) Apart from taking more repeat readings, suggest one way in which the method could be improved to collect better quality data **and** explain how the improvement would give better data. [2]

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- (d) Explaining your reasoning and using an equation from page 2, calculate the size of the air resistance force acting on a stack of 5 cake cases when travelling at terminal speed. (Gravitational field strength, $g = 10 \text{ N/kg}$). *Space for workings.* [3]

Air resistance = N

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